MEDUSA II

3-IN / 8-OUT USB & DIN MIDI ROUTER

- 3 independent DIN MIDI inputs, freely routable to 8 DIN outputs
- USB Host MIDI input with its own independent output routing
- Per-input filters for Clock, Active Sensing, Aftertouch & channel
- USB channel duplication for driving multiple parts at once
- OLED display, rotary encoder control, EEPROM setting recall

Overview

What MEDUSA II does and how it's laid out

MEDUSA II is a MIDI routing hub built around a Teensy 4.1. It takes in three DIN MIDI inputs (A, B, C) and a USB Host MIDI connection, and can send any combination of those sources out across eight independent DIN MIDI outputs. Routing, filtering and channel behaviour are all set from the front panel and stored permanently in EEPROM, so the unit remembers its configuration after power-off.

WHAT IT'S FOR

- Merging and splitting DIN gear across a rack — send any input to any combination of outputs.
- Bringing a USB MIDI keyboard or controller into an all-DIN studio, with its own routing table.
- Cleaning up a MIDI chain by filtering out Clock, Active Sensing or Aftertouch per input.
- Isolating problem channels, or duplicating a USB part across several channels at once.

CONNECTIONS

Inputs A / B / C

(rear, 5-pin DIN)

Standard MIDI DIN inputs. Each has its own filter settings and can be routed to any of the 8 outputs independently of the other two.

Outputs 1-8

(rear, 5-pin DIN)

Each output receives the merged, filtered stream of every input currently routed to it.

USB Host port

Plug a class-compliant USB MIDI device directly in — no computer needed. Routed and filtered separately from A/B/C via its own screens (4 and 5).

Front Panel Controls

Three controls drive the entire unit

ENCODER (rotary + push)

Turn

Move the cursor around the current screen's grid.

Short press

Toggle the selected square on/off (routing, filter or channel).

Long press (2s)

Save the current configuration to EEPROM.

SCREEN BUTTON

Short press

Cycle through the DIN screens: Routing -> Filters -> Ch 1-8 -> Ch 9-16.

Long press (2s)

Panic / All-Notes-Off — sends CC123 on every channel to every output.

CONFIG BUTTON

Short press

Jump to the USB screens (Output Routing / Channel Duplicate) and back.

Long press (2s)

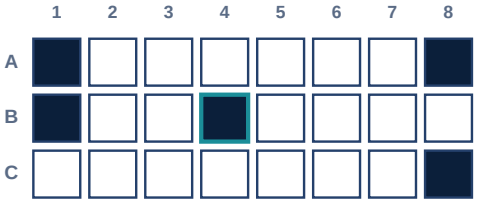
Toggle the OLED display on or off (settings keep running either way).

DIN Screens

Routing, filters and per-channel control for inputs A / B / C

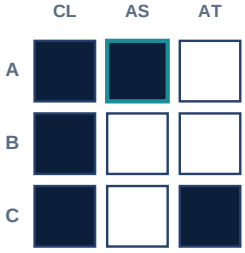
1. DIN ROUTING MATRIX

A 3x8 grid — rows A, B, C against outputs 1-8. A filled square means that input is routed to that output. Any input can feed any number of outputs at once.



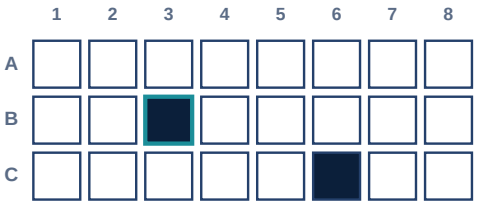
2. INPUT FILTERS

A 3x3 grid — rows A, B, C against Clock (CL), Active Sensing (AS) and Aftertouch (AT). A filled square means that message type is blocked from that input before it reaches any output.



3. CHANNEL FILTERS (1-8 and 9-16)

Two more 3x8 grids, one for channels 1-8 and one for 9-16, again by input A/B/C. Filling a square blocks that MIDI channel on that input. Turning a channel filter ON sends an immediate All-Notes-Off for that channel, so re-enabling filtering never leaves hanging notes.



USB Screens

Routing and channel handling for the USB Host input

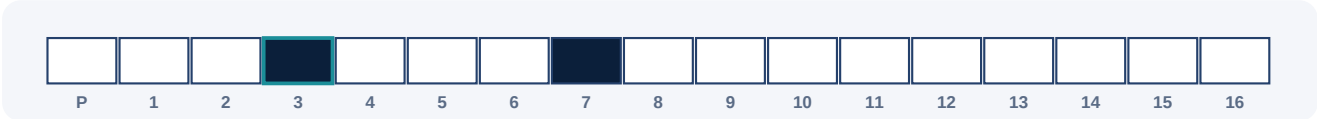
4. USB OUTPUT ROUTING

A single row of 8 squares — one per output. A filled square means the USB Host input is sent to that output. This is independent of the A/B/C routing matrix, so USB gear can feed a different set of outputs to your DIN inputs.



5. USB CHANNEL DUPLICATE

17 squares: P (Pass) plus channels 1-16. With P selected, USB MIDI passes through unchanged on its original channel. Selecting one or more numbered channels instead duplicates every incoming USB message onto all of those channels simultaneously — useful for triggering several multitimbral parts from one USB controller. Changing the selection automatically sends All-Notes-Off to every USB-routed output, so switching mid-performance won't leave stuck notes.



Saving Settings & Troubleshooting

Getting your configuration to stick, and fixing the odd hiccup

SAVING & RECALL

- Every change (routing, filters, channels, USB settings) takes effect immediately.
- Nothing is written to EEPROM until you long-press the encoder (2s) — you'll see a SAVED! message.
- On power-up, MEDUSA II automatically loads the last saved configuration.
- If no valid save is found (first boot, or after a firmware update that changes the layout), it starts with a safe default: each input routed to Output 1 and Output 8 only.

PANIC / ALL-NOTES-OFF

Long-press the SCREEN button (2s) at any time to send an All-Notes-Off (CC123) on every channel to every output. Use this to clear stuck notes without power-cycling the unit or touching your routing.

TROUBLESHOOTING

No display on power-up

Check the CONFIG button wasn't held during boot; long-press it again to toggle the screen back on.

A note gets stuck

Use the panic function above. If it recurs on one input, check whether that input's channel filter is being toggled while notes are held.

USB device not detected

Confirm it's class-compliant MIDI and plugged into the USB Host port (not the programming/serial port), and re-seat the cable.

Settings don't survive a power cycle

You must long-press the encoder to save — routing and filter changes are live but not written to EEPROM until then.

Heavy MIDI traffic feels sluggish on screen

This is expected: MEDUSA II deliberately deprioritises the display under high MIDI load to protect timing. MIDI throughput is not affected.

Firmware: MEDUSA II v12G · Platform: Teensy 4.1

Mindburner MIDI Products · Belfast

Updating the Firmware

Loading new firmware onto MEDUSA II

MEDUSA II's firmware can be updated at any time using the browser-based MINDBURNER Teensy Loader — no software install required. The same link is also available from the Mindburner website, under Firmware Updates.

Loader: mindburners.github.io/Teensy-Loader-MINDBURNER

Also linked from mindburnermidi.com · Firmware Updates

UPDATE PROCEDURE

A software-triggered reboot isn't fully reliable on MEDUSA II because the USB Host controller can still be active when a reboot is requested, so the Teensy's physical Program button is the dependable way in — the loader page will prompt for this automatically.

1. Click the button on the loader page.
2. Unplug all cables from MEDUSA II, including DC power.
3. Remove the top cover (or reach underneath with a flat tool).
4. Hold the small Program button on the green Teensy board.
5. Plug in a USB cable while still holding the button.
6. Release the button.
7. Choose the port from the popup (usually shown as an unknown device).

The update starts automatically once the port is selected — no further action needed.

